Determining hysteresis, remanence and coercive fields of a ferromagnet

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In this paper different ferromagnetic core materials were used to plot their hysteresis curves and determine their magnetic behavior. With those curves it was possible to identify characteristic properties like coercivity, saturation magnetisation, core loss and remanence. For PERMENORM 5000Z which is a nickel-iron alloying such as the following PERMENORM 5000H2: the magnetisation saturation is $B_{sat,5000H2} = (1.4441 \pm 0.0079)\,\mathrm{T}$ and $B_{sat,5000Z} = (1.428 \pm 0.066)\,\mathrm{T}$, the remanence $B_{rem,5000H2} = (0.6603 \pm 0.0034)\,\mathrm{T}$ and $B_{rem,5000Z} = (1.428 \pm 0.066)\,\mathrm{T}$ and the coercivity $H_{coer,5000H2} = (3.63 \pm 0.29)\,\frac{A}{m}$ and $H_{coer,5000Z} = (14.6 \pm 1.1)\,\frac{A}{m}$. The core loss was not measurable for the PERMENORM 5000Z in contrast to the core loss $\xi_{cf} = (2.62 \pm 0,45)\frac{J}{kg}$ for the mentioned PERMENORM 5000H2. Additionally the remanence, coercitivity and core loss result in $B_{rem,ST37K} = (0.547 \pm 0.012)\,\mathrm{T}$, $H_{coer,ST37K} = (358 \pm 14)\,\frac{A}{m}$ and $\xi_{cf,ST37K} = (0.264 \pm 0.045)\,\mathrm{T}$. It was not possible to calculate the saturation magnetisation because of the limited effective range. Lastly ferromagnets are compared with a non magnetic material as wood. With this material a linear relation of B and H was recorded. For wood the permeability was calculated $\mu_r = 1.835 \pm 0.024$.

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1 Introduction

Ferromagnets are very important nowadays as they have a memory contrary to para- and diamagnets. Thus they are used for electromotors of cars or other machines, CDs or harddiscs. They are utilized as datastore because of their characteristics as remanence and coercitivity. Their hysteresis curve does not follow a function of B(H) like dia- and paramagnets as it depends on the state it was in before (magnetised or not) and the magnetic field, which effects the ferromagnet. Sizes like remanence, coercive fields and saturation magnetisation are specific for the material. Hence we determine sizes like those by hysteresis curves in the following paper and compare them to a non-magnetic material as wood.

2 Theory

First of all it is important to clear some terms and definitions in relation to the theory of these following experiments. (As before mentioned ferromagnets are used to save information in long-term like CD´s. But there are also other types of magnets, which can´t be used for saving data for a longer period of time. To compare some attributes of ferromagnets and one of the other formes named diamagnets, wood is used as an example of diamagnetism. Diamagnets, other than ferromagnets are not able to construct a magnetic field without an outer magnetic induction. They create an inner magnetic field antiparallel to the outter one. As wood is only slightly diamagnetic, this effect cannot be observed properly. So if there is an exterior magnetic field which tries to put information inside the diamagnet, all of these pieces of information are gone when the outer field is turned off because of the material structure.) Ferromagnets are able to build their own magnetic field without an exterior magnetic induction as mentioned before. That´s because of the material structure of ferromagnets. The magnetic moments on the inside of an atom tend to align parallel. So if there is an outer magentic field the moments inside the magnet orient themselves towards the direction of the exterior one. This happens in a macroscopic scale. In a minor dimension,the microscopic one, so called magnetic domains direct themselves. Magnetic domains are regions in a magnetic material where the spins of electrons couple with each other and point to the same direction. While the outside magnetic field gets stronger, these domains get bigger and their borders migrate. So more and more

magnetic moments allign themselves like the tendency of the outer field. These borders are called domain walls. If there is no outside influence there are different materialspecific magnetic domains in one substance and the domain walls are steady which influence characteristic sizes like remanence, coercitivity and core loss. So if the outer magnetic field H increases and therefore the domain walls expand in the ferromagnet, the magnetic field B increases. This relation B(H) is shown in so called hysteresis curves. Regarding to this curves there are a few terms, which should be explained beforehand. If the voltage range is big enough, the saturation magnetisation can be observed. If the saturation magnetisation occours, all magnetic domains are alligned parallel to the outer field. When hitting this spot the magnetic field B is only increasing with the incline of the vacuum permeability μ_0. If now the outer magnetic field is reduced constantly some of the magnetic domains start to change their orientation because of thermal fluctations and the magnetic flield decreases. This happens on a different trajectory than before because the thermal fluctation is less than the spin interaction in ferromagnetic materials with room-temperatur. So if the point $H = 0$ is passed there is magnetism left, which is called remanence. After that the direction of the outer magnetic field is inverted and enlarged. Eventually the curve gets to the spot where the magnetic flux is completely gone. The value of H at this point is called coercitivity. Now the procedure is the same as before but with a reversed magnetic field. The hysteresis curve is complete if the saturation magnetization from the beginning is achieved again. So every hysteresis curve is point-symmetrical. When the magnetic domains shift in a different direction, energy is lost in form of heat. This heat equals the area inside the hysteresis curve and can be calculated with the formula:

$$\xi = \frac{E}{m} = \frac{1}{\rho} \oint B dH \tag{1}$$

3 Experiment

In the following experiments hysteresis curves of different materials were taken with the experiment set-up given in fig. [1]. The installation consists of two toroidal inductors, one with the material to be analysed inside of it. This one generates the outer magnetic field which induces a voltage and electricity in the induction coil. The generated magnetic field strength H is given by the voltage drop U_R measured with a known resistance R with the formula

$$H = \frac{N_1 U_R}{\pi \frac{1}{2} (d_i + d_a) R} \tag{2}$$

with N_1 the winding number, d_i and d_a the inner and

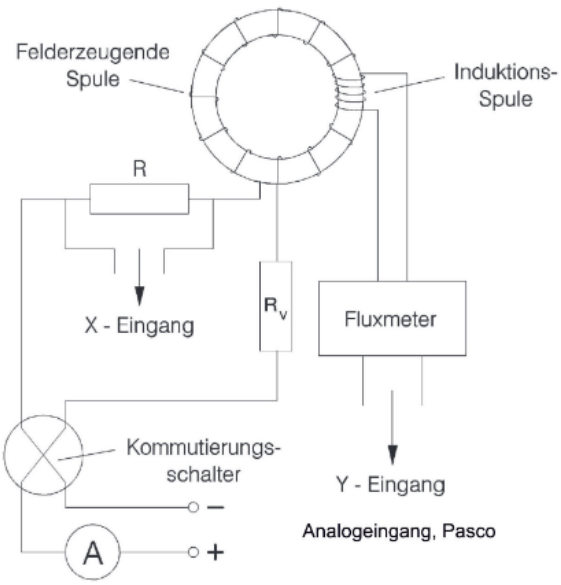

Fig. 1: Build up of the following experiment to plot hysteresis curves of different materials. The basic construction stays the same only the two given resistances R_v and R and the material of the core change in some sections.

outer diameter of the field creating inductor. A resistance with the value $R = (66,7 \pm 1,4)\,\Omega$ is used for measuring low electricity . Whereas for higher electricity the measuring resistor $R_m = 0.20\,\Omega$ is utilized. With the change of the flux density a voltage surge in the induction coil is created. Via the built in fluxmeter, this voltage surge can be measured against time to get a tension which is proportional to the flux density B calculated by the formula:

$$B = \frac{\gamma_f U_f}{N_2 A_{eff}} \tag{3}$$

Where γ_f is an adjustable proportionality factor of U_f. N_2 is the winding number of the induction coil. A_{eff} the area which is actual saturated by the magnetic field and measured by

$$A_{eff} = \frac{1}{2}(d_a - d_i)\, h\, f \tag{4}$$

with the diameters d_a and d_i, the hight h and the bulk factor f of the core. By these means it is possible to plot the hysteresis curves with the Interface called PASCO Universal Interface 550 Analog-Digital-Wandler. However, because of the functionality of the fluxmeter and the integrated operational amplifiers all values are shifted in the direction of the flux density. To get the right plots these data points have to be centered with:

$$U_c = U_B - \frac{1}{2}(U_{min} + U_{max}) \tag{5}$$

Whereas U_c is the centered voltage of the induction coil and U_B the induced voltage in it. U_{min} and U_{max} are the lowest and highest measured voltages. Because there is no infinite input resistence plugged in the fluxmeter there is also a linear shift, rising by time linearly, regarding to the data points of B. How this disruptive factor is adjusted is explained in 5.1 on the basis of the hysteresis curve of wood.

3.1 Measurement with the core PERMENORM 5000H2

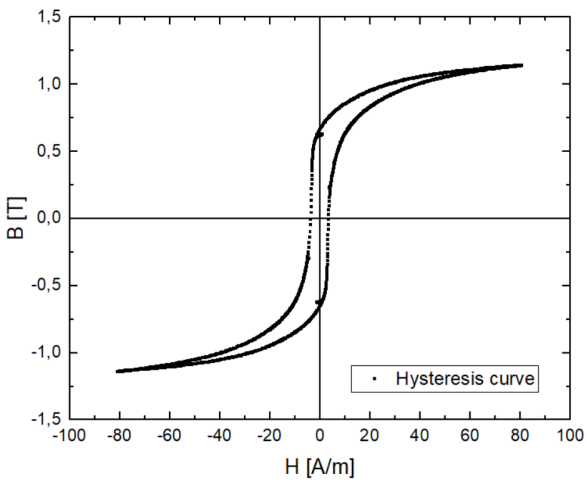

Fig. 2: Hysteresis curve of the alloying PERMENORM 5000H2 used for calculating the values of remanence, coercivity and core loss.

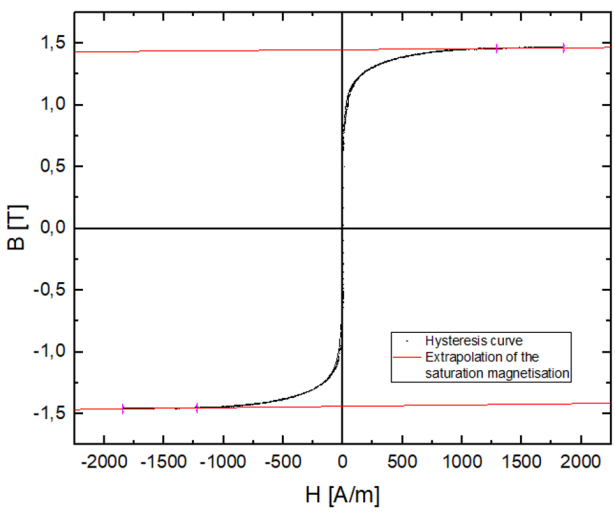

Fig. 3: Hysteresis curve of PERMENORM 5000H2 used with extended effective range to determine the saturation magnetisation which is calculated with the help of extrapolation.

In this first experiment the hysteresis curve of the core PERMENORM 5000H2 is plotted. The material has an iron-nickel alloying. The resistance $R = 66.80 \pm 0.34\,\Omega$ is used to regulate the electrical current below $I_{max} = 100\,\text{mA}$. The procedure to receive the hysteresis curve is described in sec. 3. The curve is not closed-formed regarding to disruptive factors, but can be corrected afterwards, which is explained in 5.1. The recieved and corrected hysteresis curve is shown in figure [2]. By means of the hysteresis curve, characteristic sizes like remanence and coercitivity can be determined. To obtain those, a linear fit is placed at the intercepts, shown in fig. [4]. The data is put in chart [1]. Since the voltage range, creating the magnetic field strength H, is too narrow to measure the saturation voltage, the hysteresis curve is recorded with a current range of $I_{max} = 2.3\text{V}$. Therefore, a resistance of $R = 0.2\,\Omega$ is applied. The resistance is treated without an uncertainty as its one is much smaller than the other ones and therefore negligible. The resultant curve is shown in fig. [3]. The saturation magnetisation is ascertained by the help of a linear fit and their intercept with the y-axes plotting the magnetic field B. The resulting sizes are portrayed in tab. [1].

References regarding saturation magnetization and coercivity state the value of these characteristics for PERMENORM 5000H2: $B_{saturation,Lit} = 1.55$ T and $H_{coercivity,Lit} = 5\ \frac{A}{m}$. In this experiement the values of the searched characteristics do not agree with literature values. This could be due to the fact that the saturation magnetisation was not reached or that the process of manufacturing has an influence on the magnetisation characteristics.
This material can be used for a transformer due to the small coercivity since it is very easy and energy-efficient to change the poles. Due to this fact this material is unsuitable for long lasting permanent magnets and their use or storing data.

$B_{rem}\,[mT]$	$H_{coer}\,[\frac{A}{m}]$	$B_{sat}\,[mT]$	$\xi_{cf}\,[\frac{mJ}{kg}]$
660.3±3.4	3.63±0.29	1,444.1±7.9	2.62±0.45

Tab. 1: Calculated values for the alloying PERMENORM 5000H2

3.2 Flat steel type ST37K

In figure [4] the received hysteresis curve of flat steel is shown. The core of the coil is of the type ST37K. Again, by the intercept points of curve and axes we get the values of remanence B_{rem} and coercivity $H_{coercivity}$. Since it was not possible to get the saturation magnetisation based on the too small voltage range creating a magnetic field strength H in

the inducter coil, there is no value measured in this task. The core loss ξ_{loss} in this experiment is calculated through numerical integration of the area in the curve with formula (1).

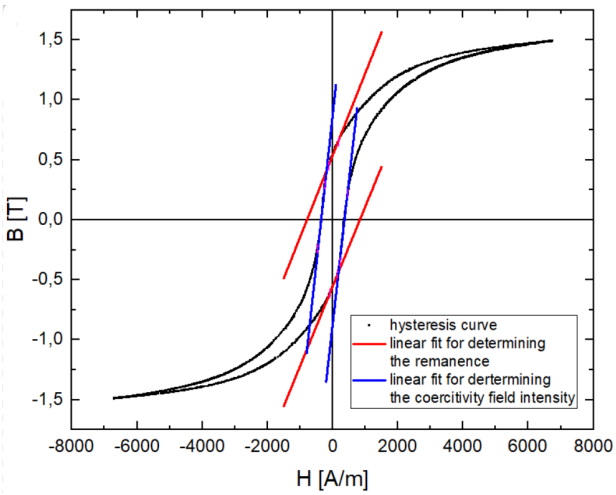

Fig. 4: Plotted hysteresis curve of a steel core of the type ST37K. Resulting characteristics of this material like remanence and coercivity were calculated by the help of the shown reference lines.

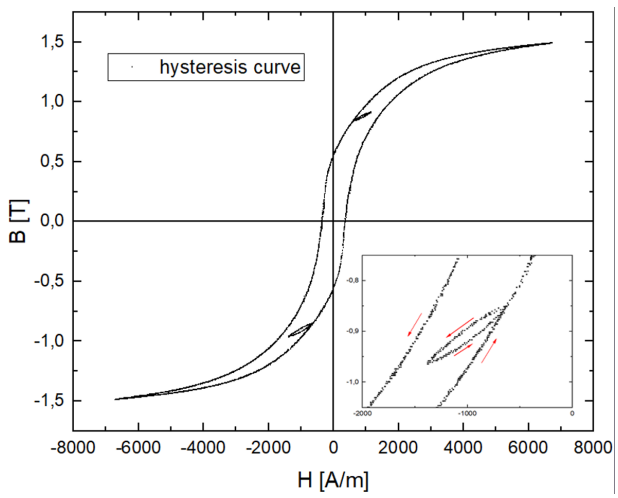

Fig. 5: Hysteresis curve of a steel core type ST37K with additional loops. One of these loops is shown in a close-up picture in the bottom-right-hand corner. The direction of the hysteresis curve is indicated with arrows.

The change of direction of the graph is analysed in addition to the normal hysteresis curve in figure [5]. Therefore the normal procedure plotting the curve was done, but after reaching the starting point a second round was started. However in this following circle in two points the magnetic field was not constantly decreased but changed for a short time to increase again by H. There are two loops shown in fig. [5]. They result by decreasing/increasing H contrary to the direction before for a short period of time. B follows now a different track than before because of the hysteresis effect explained in **??**. After that the magnetic field is now increased/decreased opposed to the direction before.

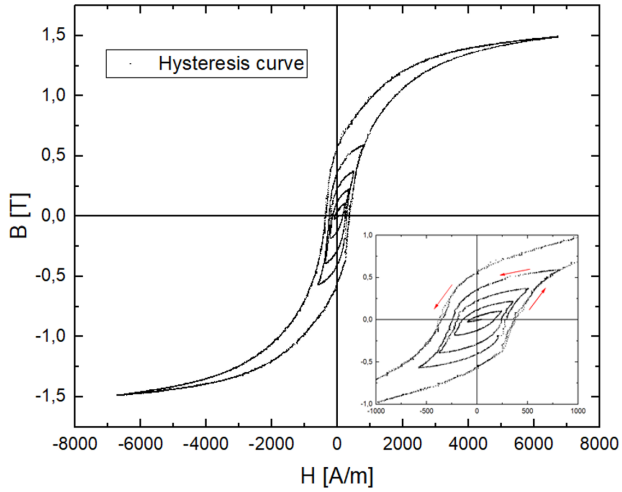

Fig. 6: Plotted hysteresis curve of steel core type ST37K. This curve shows the demagnetisation of the material in form of decreasing loops converging to zero. It is visible that the point of zero magnetisation is not reached exactly in the close-up picture of the inner area on the bottom-right-hand corner.

In fig. 6 the same effect was used by demagnetising the core: Before reaching the starting point again, the magnetig field strength H is decreased and so on approximating to zero. After this the outer magnetic field is increased until the starting point is hit again. So the resulting plot is a closed hysteresis curve with some loops inside. The actual execution was a little bit hard as the off-set had to be considered and the shift. Hence a small remanence is remaining. But comparable with the earlier determined remanence in fig. [4] this one is way smaller which is indicating that this method works very well. Due to the high coercive field, the steel core ST37K is not so well used for transistors but storing data.

B_{rem} $[T]$	H_{coer} $[\frac{A}{m}]$	B_{sat} $[T]$	ξ_{cf} $[\frac{J}{kg}]$
0.547 ±0.012	358 ±14	-	0.264±0.045

Tab. 2: Calculated values for the material flat steel of the type ST37K

3.3 Measurement with the core PERMENORM 5000Z

PERMENORM 5000Z is another iron- nickel alloying. The instruction to get the characteristics is identical to the other core with an alloying. The values of remanence and saturation magnetisation are given in table [3]. Because there is no given data for the density of this material it is not possible to calculate ξ_{loss}.
Like PERMENORM 5000H2 this core with the alloying 5000Z can also be classified as a magnetic soft material. Which is used for short time memory in transistors because of the small coercivity and core loss as described before.

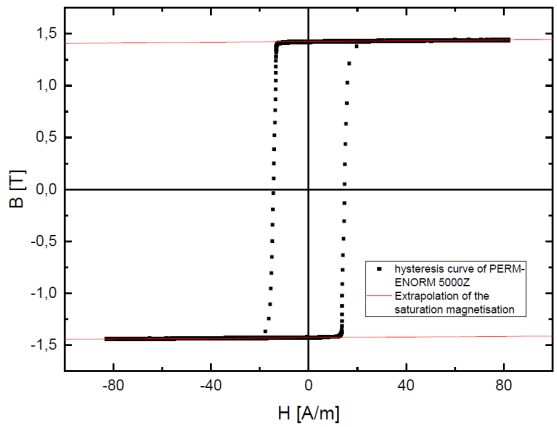

Fig. 7: Hysteresis curve of PERMENORM 5000Z als alloying. Out of this plot all values of the material are caluclated because of the visible area of saturation magnetisation which is nearly the value of the left remanence.

B_{rem} $[T]$	H_{coer} $[\frac{A}{m}]$	B_{sat} $[T]$	ξ_{cf} $[\frac{J}{kg}]$
1.428 ±0.066	14.6 ±1.1	1.428 ±0.066	-

Tab. 3: Calculated values for the alloying PERMENORM 5000Z

3.4 Wooden core for comparison

The course of action is in these cases always the same. So the hysteresis curve of wood is plotted and shown in figure [8]. This material was selected to be able to compare a ferromagnetic substance with a non ferromagnetic, slightly diamagnetic one. In the image the linear correlation of B and H is nicely illustrated. Also it is recognisable that the inner magnetic field is diminutive because wood is not magnetic so there is no inducting field possible. This is due to the inner structure of wood. The connection of the atom spin do not connect with each other as strong as in ferromagnets so it is impossible to bulid up magnetic domains. Because these bonds are so weak the termic fluctation is relatively strong and destroys any parallelism after decreasing the outer field. Therefore, there is no hysteresis curve nor a left remanence at $H = 0\,\frac{A}{m}$. The given gradient of the curve is exactly the permeability $\mu = \mu_0 * \mu_r$. We receive the permeability of wood $\mu_r = 1,835 \pm 0,024$ with $\mu_0 = 1,25663706212(19) * 10^{-6}\,\frac{N}{A^2}$. According to this result wood must be paramagnetic. Obviously this cannot be true. Wood is a diamagnetic material with $\mu_r < 1$ as mentioned before. This must be down to the experiment set-up as all the coils with their inner core are stacked in a shelf right next to each other. So we assume that the ferromagnetic cores above and below sophisticate the permeability μ_r of wood. Moreover the shelf is made out of steel which contributes to the distortion.

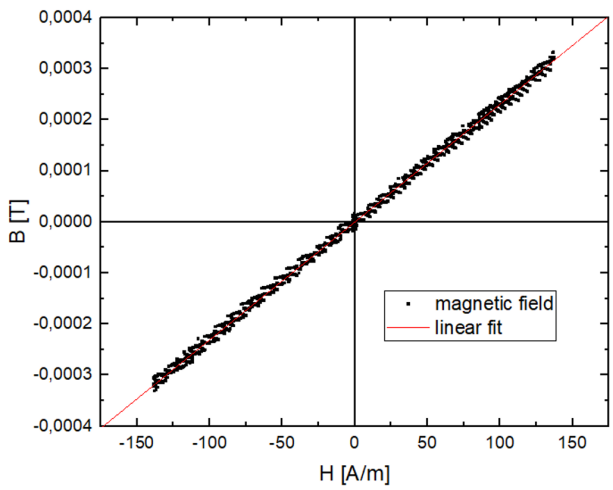

Fig. 8: Plot of a wooden core which shows a linear behavior and no hysteresis curve.

4 Results

In this paper three different ferromagnets and one theoretical diamagnet were analysed. With the help of plotted hysteresis curves it was possible to calculate characteristics like remanence, saturation magnetisation and some other ones which are all specific for those materials. To be able to compare the behaviour of ferromagnets with non magnetical substances: wood was the material of choice. Now all values of the individual cores are listed beneath: For the ST37K steel core we got for remanence $B_{rem} = (0.547 \pm 0.0012)$T, for coercivity $(385 \pm 14\,\frac{A}{m})$ and the core loss with the value $\xi_{cf} = (0.264 \pm 0.045)\,\frac{J}{kg}$. For the two iron-nickel cores we got for PERMENORM 5000Z the re-

manence $B_{rem} = (1.428 \pm 0.066)\ T$, the coercivity $H_{coer} = (14.6 \pm 1.1)\ \frac{A}{m}$ and the saturation magnetisation with the value $B_{sat}(1.428 \pm 0.066)\ T$. For the other one with the allying PERMENORM 5000H2 we got $B_{rem} = (660.3 \pm 3.4)$T, $H_{coer} = (14.6 \pm 1.1)\ \frac{A}{m}$, $B_{sat}(1,444.1 \pm 7.9)\ T$ and at least $\xi_{ef} = (2.62 \pm 0.45)\ \frac{mJ}{kg}$. To refer to the wooden core which shows a linear behavior it was possible to measure the permeability $\mu_r = (1.835 \pm 0.024)$.
All curves are corresponding to the theoretical expectations. Additionally the turn-around in the hysteresis curve was detected with the steel ST37K core. With this technique it was possible to demagnetise the mentioned core and to plot the received curve.

Literatur

[1] Meschede, Gerthsen Physik, 23, 2005

[2] Frauenfelder und Huber, Einführung in die Physik II,

[3] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 39

[4] https://vacuumschmelze.com/products/soft-magnetic-materials-and-stamped-parts/40-to-50-Nickel-Iron

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5 Anhang

5.1 Drift of the fluxmeter

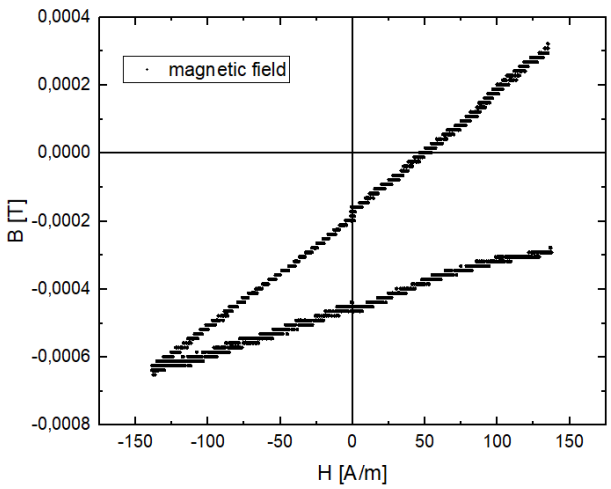

Fig. 9

In figure 9 the unadjusted plot of the wooden core is represented. The B- offset is aligned but not the drift of the fluxmeter. As you can probably tell the flux density is drifting towards lower values with a seemingly constant speed. Wherefore the shift is assumed to be linear. Hence the final plot is not closed and does not hit zero. With the following formula it is possible to calculate the drift of the fluxmeter for each datapoint and hereby adjust the plot correctly

$$\dot{B}_{drift} = (\frac{\partial B}{\partial t})_{drift} \approx \frac{B_n - B_1}{t_{total}} \tag{6}$$

by the use of the first recorded flux density B_1 and the last one B_n and the duration of the whole testseries t_{total}. So this received time shift was added to all datapoints to get the adjusted final plot.

5.2 Troubleshooting

The given values at the workplace were supposed to be accurate and without error. So it is possible to calculate σ_B and σ_H with gaussian error propagation and the formulas [2] and [3].
The error of the fluxmeter is estimatet because the manufacturer information was missing. We set 0.5% as the value of the error. For other used devices Voltcraft 333 and the interface: we took the errors of the manufacturer's data $\sigma_{U,interface} = 1.22\,\text{mV}$ and $\sigma_{VC333} = \pm(0.5\% + 2\text{digits})$.

5.2.1 Uncertainty of remanence and coercitivity

As there is an uncertainty in H and B there is like a uncertainty square around every data point. This is why we calculated the uncertainty for the remanence and coercitivity by multiplying one error with the associated tangent slope comparing those in e.g. fig [4]:

$$\sigma_{rem} = \sigma_B + \sigma_H\, m_{rem} \tag{7}$$

and:

$$\sigma_{coer} = \sigma_H + \frac{\sigma_B}{m_{coer}} \tag{8}$$

Here the tangent slope m_rem portrays the slope average of the tangents at the remanence. Analogous it is apllied for m_coerc.

5.2.2 Uncertainty of the saturation magnetisation

As the saturation magnetisation is determined by applying a tangent to the saturation, the uncertainty is calculated by linear regression.

5.2.3 Uncertainty of the resetting

Since the resetting is calculated by origin integrating the area of the hysteresis curve, we used the average of the relative uncertainty of H and B and accounted them by gauss.

5.3 coil data

	Fe-Ni H2	Fe-Ni Z	flat steel	wood
d_a	260mm	260mm	260mm	260mm
d_i	220mm	220mm	220mm	220mm
N1	605	610	1010	1010
N2	80	56	53	250
h	30,0mm	30,0mm	30,0mm	30,0mm
f	90%	90%	100%	100%
R	0,20Ω	0,20Ω	0,20Ω	0,20Ω

Tab. 4: values of the different coils used in this experiment